

Robots in Context

Lecture 4

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Agends

- Programming exercise 2 presentations:
 - Mads
 - Jakob (next week)
- Lecture 3
 - Feature extraction, PE3 volunteers?
- Lecture 4
 - Configuration / work-space
 - Potential fields
 - Roadmaps



Lecture 4

- Programming Exercise 3
Volunteers:

1.

2.



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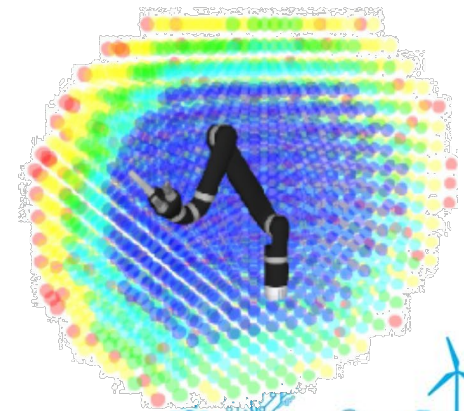
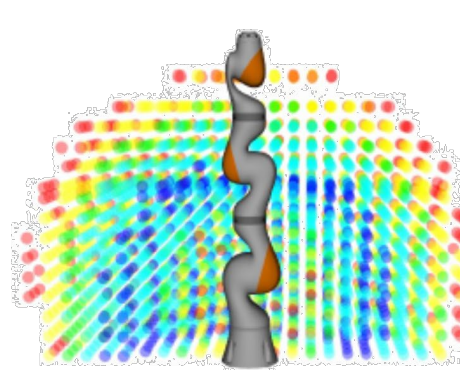
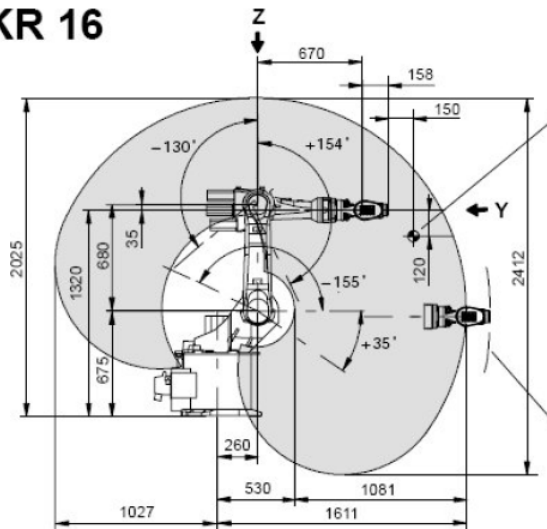




Work Space

- The space in which the robot works
 - The set of points that are reachable by the robot
- Typically represented by Cartesian Coordinates

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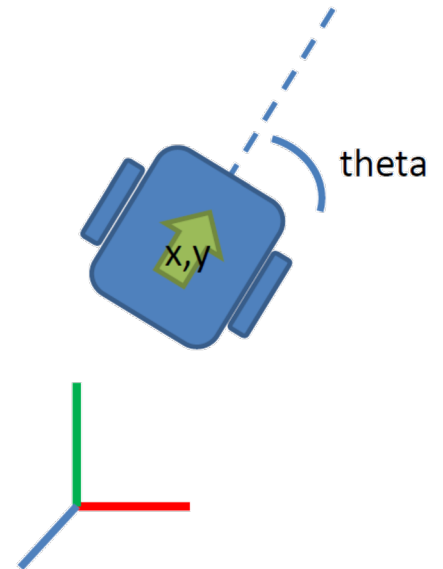
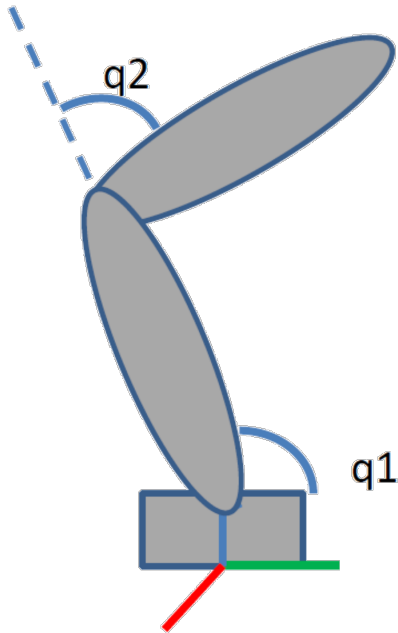


[Makhal & Goins, Reuleaux: Robot Base Placement by Reachability Analysis]



Configurations

The **configuration** describes the *relevant* state of the system and enables us to compute every point in the system.



Configuration Spaces

- The space of all possible configurations of a robot.
 - Sometimes denoted C-space

- Consists of free and occupied space

$$C = C_{free} \cup C_{occupied}$$

- Can be high-dimensional
- Bounded vs. Unbounded
- In some applications the configuration space could also include velocities, accelerations and other relevant information.





- Add up DoF per joint (R/P/S)

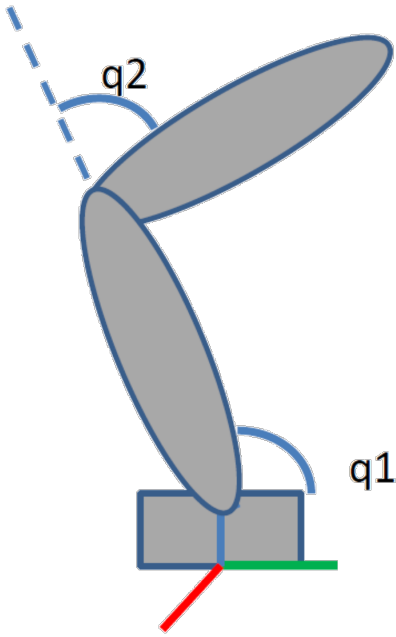


$$M = N(k - n - 1) + \sum_{i=1}^n f_i$$



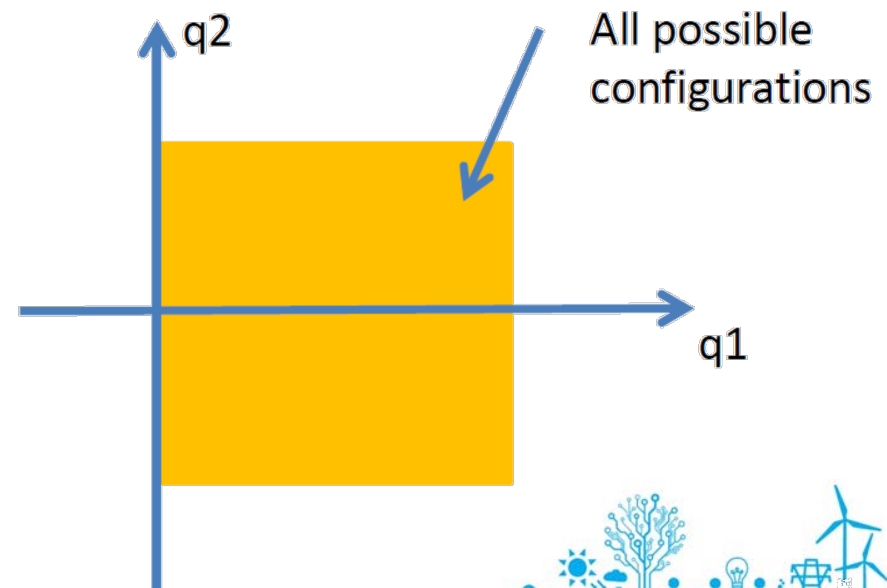
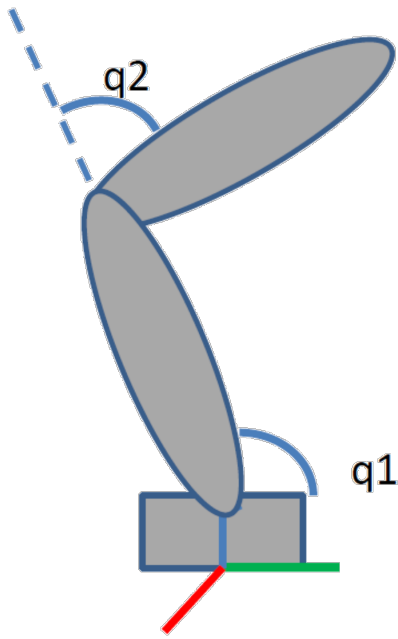
Configuration Space

- Describe the configuration space for this robot arm?



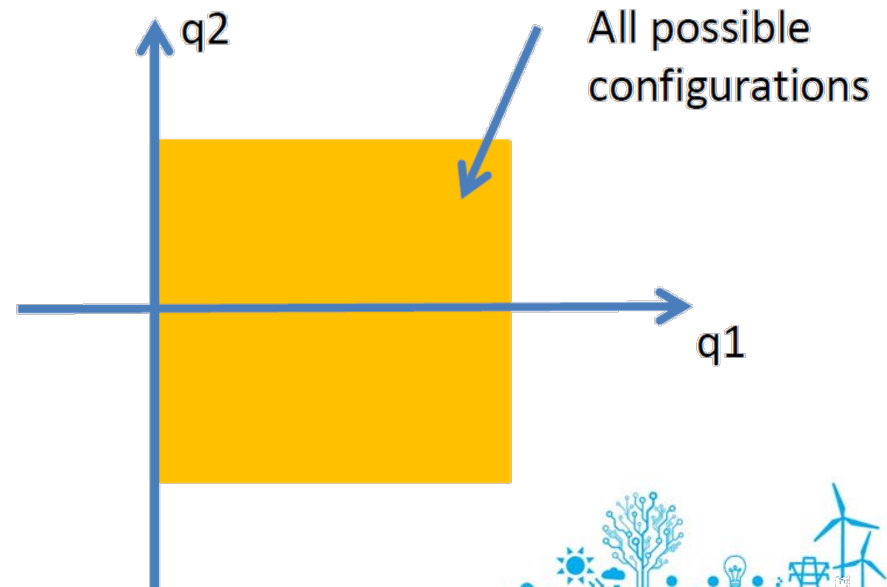
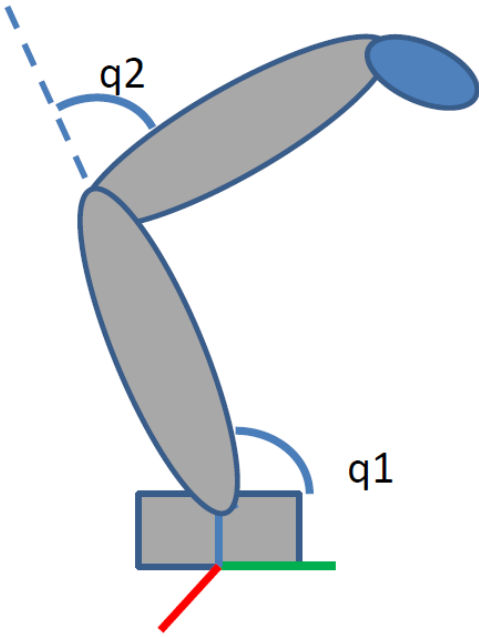
Configuration Space

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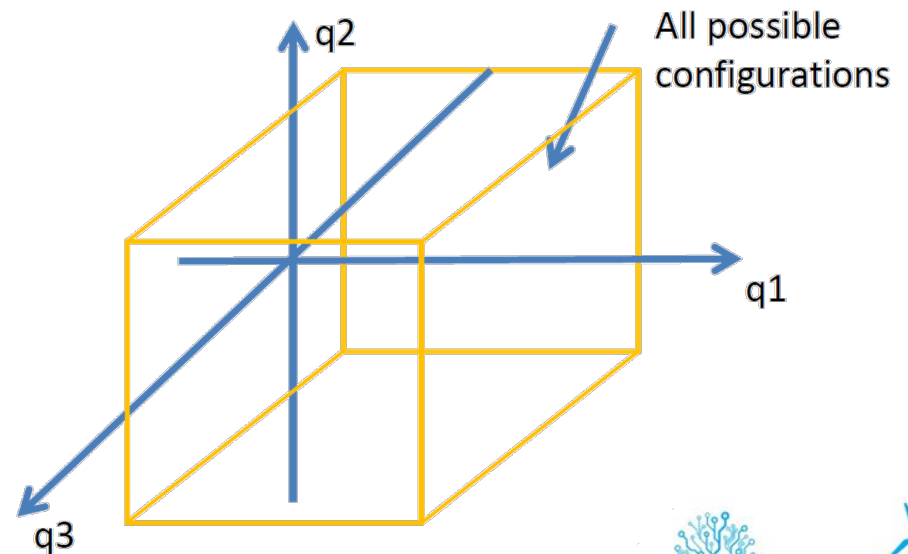
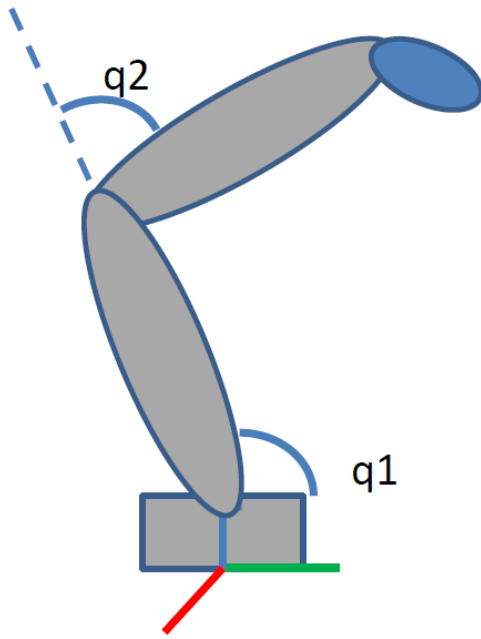
Configuration Space

- Describe the configuration space for this robot arm? What if we add one additional joint?



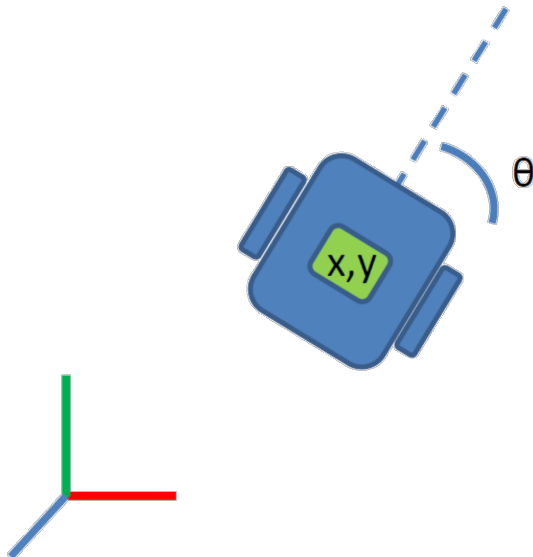
Configuration Space

- Describe the configuration space for this robot arm? What if we add one additional joint?



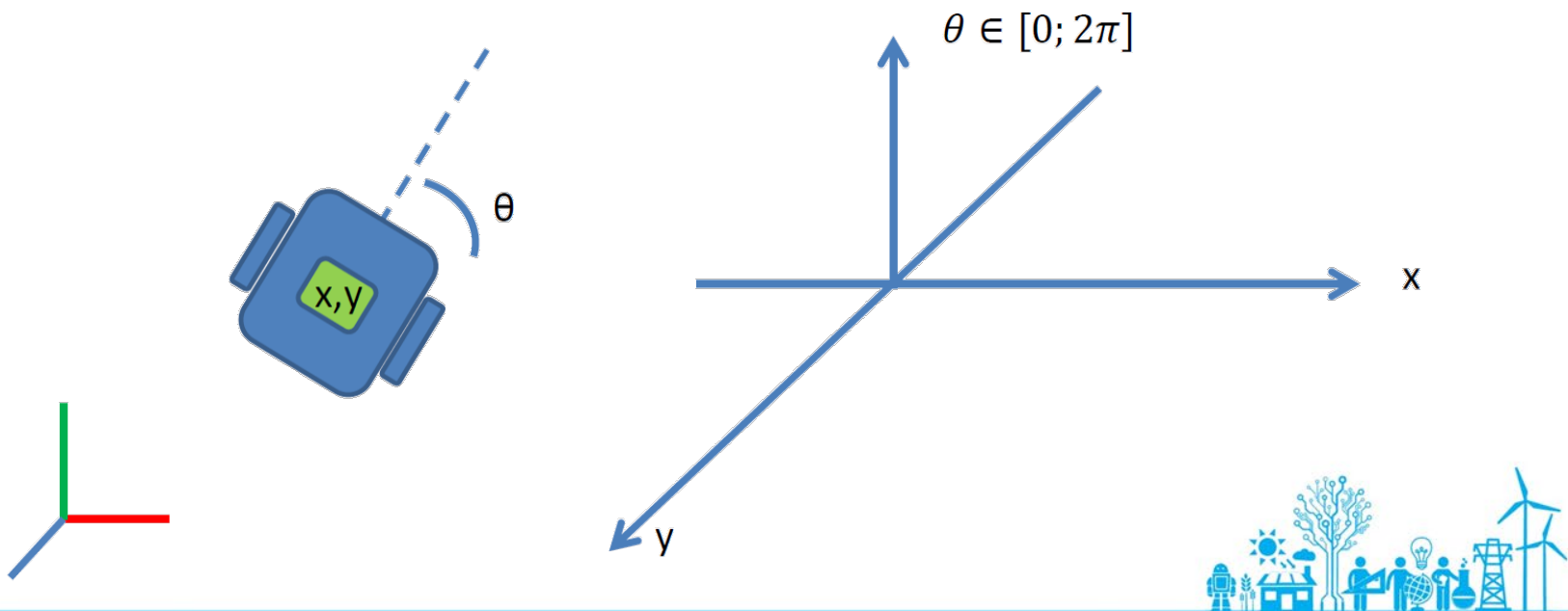
Configuration Space

- Describe the configuration space for this mobile robot?



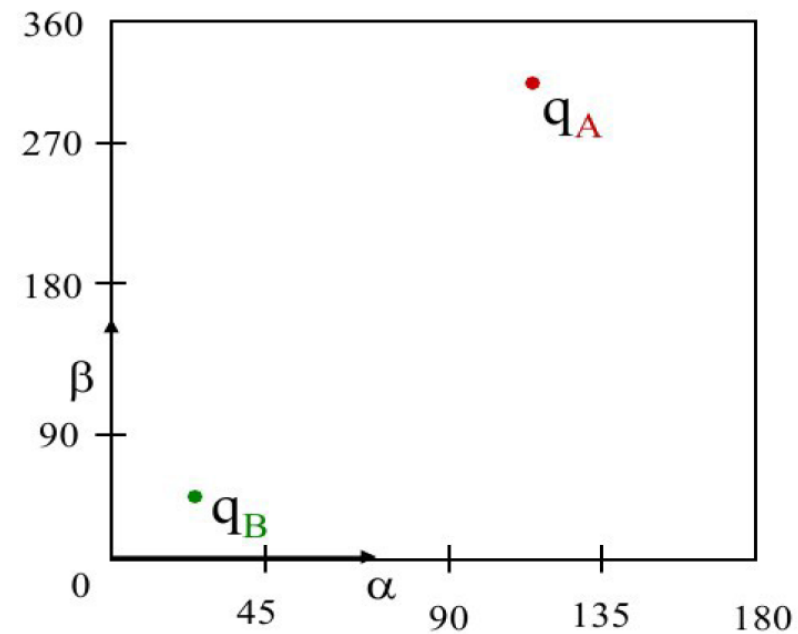
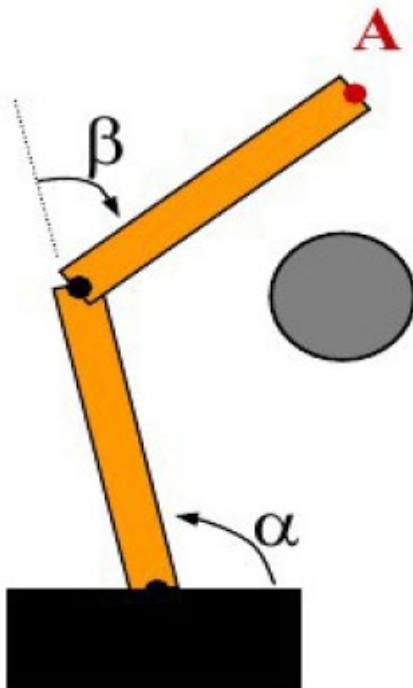
Configuration Space

- How is the configuration space for this mobile robot?

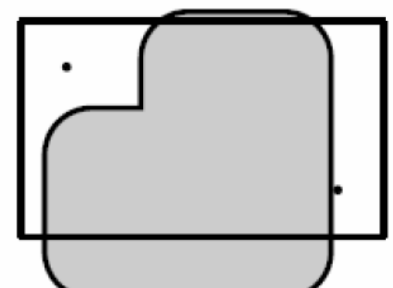
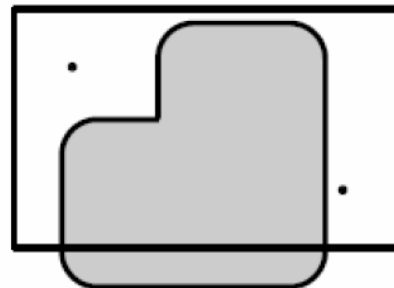
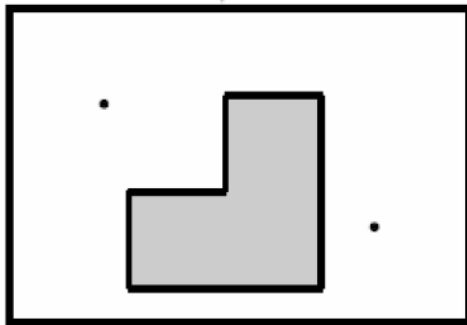


Configuration Space

Where do we put  ?



workspace



C-space



Obstacles in C-space

- Given
 - a configuration space Q
 - a work space obstacle WO_i
- Configuration space obstacle
$$QO_i = \{q \in Q | R(q) \cap WO_i \neq \emptyset\}$$
- The free configuration space
$$Q_{free} = Q \setminus (\cup_i QO_i)$$



Summary

- Workspace
 - The space which the robot operates in and that the robot can actually reach
- Configuration space
 - Often the same dimension as DoF
- Degree of freedom (DoF)
 - Number of parameters needed to define the state of system

